

PARTHIV A DHOLARIA

parthiv.dholariya9221@gmail.com | +91-9313054893 | [GITHUB](#) | [LINKEDIN](#)

Education

B.Tech , Indraprastha Institute of Information Technology, Delhi

2021 - 2025

- Computer Science and Engineering | CGPA: 8.42

Bhavans Public School , Doha , Qatar

2020 - 2021

- AISSCE (Class XII), Percentage: 94.2 %

Skills

C++ | C | Java | Python | Machine Learning | Deep Learning | CV | NLP | HTML | CSS | JS | MySQL | Django | Git | Mongo | Docker | AWS | NSSM | Linux | OpenCV | Pytorch | HuggingFace | LLMs | GenAI | RAG

Work Experience

Ripik.AI | ML Engineer

May'25 - Present

- Architected ML pipelines for CV detection, segmentation, tracking, and counting use cases, **accelerating deployment efficiency by 30%** via server hardening, atomic MongoDB read/update/write operations , storing aws s3 urls instead of base64 compressed images into the MongoDB.
- Trained YOLO models for detection and segmentation; leveraged GDINO & LangSam wrappers to engineer robust CV solutions, **increasing model accuracy by 25%**.
- Optimized Ripik Optimus to enhance production planning and manpower scheduling for Sunpharma Pharmaceuticals and Sai Life LTD, **reducing scheduling time by 20%** using novel linear time complexity algorithms.

NX Block Trades Pvt. Ltd. | ML Intern

Jan'25 - May'25

- Worked upon high frequency and volume of monthly options data and analysed options strategies.
- Engineered and backtested trading strategies, implementing quantile regression to generate ON/OFF signals, resulting in a 10% increase in cumulative profit.

CISPA-Helmholtz center for Information Security , Germany | Summer Intern

May'24 - Jul'24

- Investigated vulnerabilities in TLS handshakes and developed a machine learning system to detect attacks on servers, **increasing detection rate by 18%**.
- Automated attack simulations on diverse servers and constructed a pipeline for dataset curation, training, and evaluation.

Projects

NumerAi: crowd-sourced hedge fund data- science Competition | [Github](#)

Nov'24

- Predicted stock returns of unknown assets on live data with varying market features and concept drift, achieving **best correlation with target: 0.0221**.
- Implemented metric-based meta-learning approaches, including Matching and Prototypical Networks, ranking **1376/2117** in correlation and **894/2117** in MMC (as of 2024-11).

Evaluating Mathematical Reasoning of LLMs | [Github](#)

Nov'24

- Benchmarked LLMs (Mixtral-8x7b, LLama-3.1-8b, Gemma-9b) on SVAMP, GSM8k, and GSM-Hard datasets, achieving **accuracy up to 100%**.
- Developed a novel multi-agent framework with specializer and master combined with RAG; integrated ZeroShot, CoT, ReAct, RAG, and Teacher-Student models.

Textual Emotion-Cause Pair Extraction in Conversations | [Github](#)

Apr'24

- Engineered a two-step pipeline to extract emotion-cause pairs: emotion classification using a novel architecture, followed by QA-based cause identification.
- Ranked 5th in Subtask 1 CodaLab Competition for SemEval-2024 Task3, outperforming 95% of participants.

Achievements

- Received the Dean's Award for Academic Excellence for securing an overall **CGPA of 9 for my 3rd year**.
- **89.78% accuracy** on Google Speech Command Dataset(using GMM) and **59.798 %** on Human activity recognition dataset from UCL(using KNN).
- School Topper in 12th CBSE Board exam 2021-22.